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Negative Obstacle Detection in Off-Road Environments Using Sparse 3D LiDAR Point Cloud

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***Abstract*— Negative obstacle detection is crucial to ensure safe navigation of unmanned ground vehicles (UGV) in off-road environments. Unlike previous approaches developed using high-resolution LiDAR or multi-LiDAR fusion, this paper proposes a negative obstacle detection method applicable to a single low-resolution LiDAR. First, a ground segmentation method is used to segment the point cloud sampled by LiDAR into ground points and obstacle points. Then, the point cloud is serialized by projecting it into a front-view image. Next, combined with the ground segmentation results, the range view image is traversed, and the feature values of each point in the image are extracted and analyzed to obtain negative obstacle detection results. Finally, the detection results of multiple frames are fused using a Bayesian rule to improve the detection robustness. To verify the effectiveness of the proposed method, we conduct a series of experiments in several typical off-road scenarios. The experimental results show that the proposed method can accurately detect negative obstacles with high real-time performance.**

I. Introduction

In recent years, with the development of Light Detection and Ranging (LiDAR) sensor technology, LiDAR products with different characteristics have emerged and are widely used in the field of Unmanned Ground Vehicles (UGVs). According to UGV application scenarios, suitable LiDAR products can be selected for autonomous driving systems. For small, low-speed UGVs such as unmanned delivery vehicles, low-resolution LiDAR is often used for cost reduction. However, the sparse and uneven point cloud acquired by low-resolution LiDAR poses several challenges to the perception system. Many scholars have proposed several solutions in different directions of perception technology, such as obstacle detection [1][2], localization[3]，and object detection[4]. In addition to these perception techniques, negative obstacle detection is crucial for autonomous navigation of small UGVs in off-road environments.

A variety of negative obstacle detection methods have been proposed before, but most of them are designed for high-density point clouds. These methods mostly rely on the density geometric features of the point cloud for the detection of negative obstacles. However, different from negative obstacle detection in high-density point clouds, detecting negative obstacles in sparse point clouds faces the following challenges:1) The limited number of points makes it difficult to extract geometric features of negative obstacles. 2) For low-resolution LiDAR, the inclination angle between the laser emitters is large, leading to the possibility that the laser beam may not be able to scan negative obstacles.

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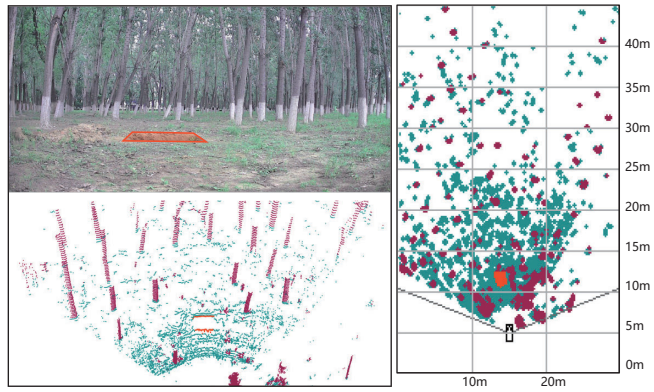

Fig 1. Negative obstacle detection in a wooded scene. The image in the upper left corner is the real-world scenario. The bottom left image shows the raw sensor data. The right image is a bird's eye view (BEV) of the obstacle detection results. The dark cyan color represents the ground. The dark red color represents the positive obstacles. The orange-red color represents the negative obstacles. The white color represents the unknown area.

To address these issues, we propose a negative obstacle detection method to improve the accuracy of negative obstacle detection in sparse point clouds, which is shown in Fig 1. The main contributions of this study are as follows

- We propose a geometric feature-based method for single-frame negative obstacle detection. Three geometric features are designed based on the LiDAR scanning model.
- A grid map-based multi-frame fusion method is proposed to improve the robustness of negative obstacle detection.
- Experiments were conducted in various typical real-world scenarios to validate the effectiveness of the proposed method.

II. Related Work

Negative obstacle detection has been a challenging research topic in the field of environmental perception technology. In order to detect obstacles below the ground, various methods based on different sensors have been proposed.

Vision-based approaches [5][6] mostly use stereo camera sensors to detect the presence of negative obstacles in the environment. This is because the depth information provided by the stereo camera effectively characterizes the road surface and facilitates further detection of negative obstacles below the road surface. However, in off-road environments, illumination variation and road surface undulation can cause

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greater disturbance to camera imaging, ultimately leading to negative obstacle detection failure.

LiDAR can accurately measure the distance of the object without being affected by light. Shang et al. [7] proposed a negative obstacle detection using multiple LiDAR sensors. In this method, two low-resolution LiDARs are mounted obliquely on each side of the UGV. Compared with the traditional upright LiDAR mounting method, this mounting method reduces the sensor detection blind zone and improves the density of the point cloud. In addition, based on the mounting method, this method designs two negative obstacle geometric features and classifies the features using a support vector machine (SVM) to identify negative obstacles. Other negative obstacle detection methods [8][9] use multiple LiDARs, which have the LiDARs mounted in much the same way as in [7], except that the geometric features of the negative obstacles are redesigned to improve the accuracy of the detection. Referring to the u-v disparity of stereo vision, Chen et al. [10] proposed a Lidar-imagery-based disparity method for constructing LiDAR-histogram and implemented a unified method for detecting positive obstacles, negative obstacles, and water body obstacles based on LiDAR histogram features. This method uses only a single high-resolution LiDAR. Li et al. [11] used one 40-channel LiDAR to detect the negative obstacle in off-road environments. This method classifies negative obstacles into two types and designs different detection features for each type to achieve obstacle detection.

III. Geometric Features of Negative Obstacles in Point Cloud

In this section, we briefly analyze the geometric features of negative obstacles in the point cloud.

Typically, a multi-layer LiDAR has a fixed number of laser emitters along the vertical axis of the sensor, each of which has a fixed inclination angle φ. These laser emitters simultaneously emit laser beams to sample the environment. At the same time, these beams synchronously rotate around the vertical axis of the sensor to obtain a fixed horizontal detection field of view. After one scan cycle, we can get a complete frame of environmental point cloud data, as shown in Fig 2.

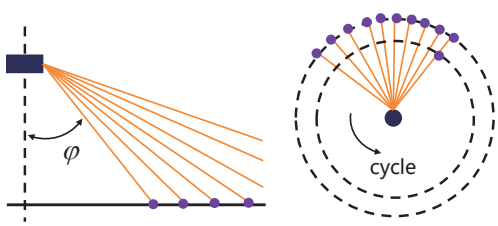

Fig 2. LiDAR scanning model in radial direction (left) and horizontal direciton (right).

In this work, the LiDAR is mounted horizontally on top of the UGV. According to the LiDAR scan configuration, each laser beam of the LiDAR gets a ring-shaped data distribution after one cycle of scanning. Since different laser beams have different inclination angles, the radius of the ring is different, as illustrated in Fig 3(b).

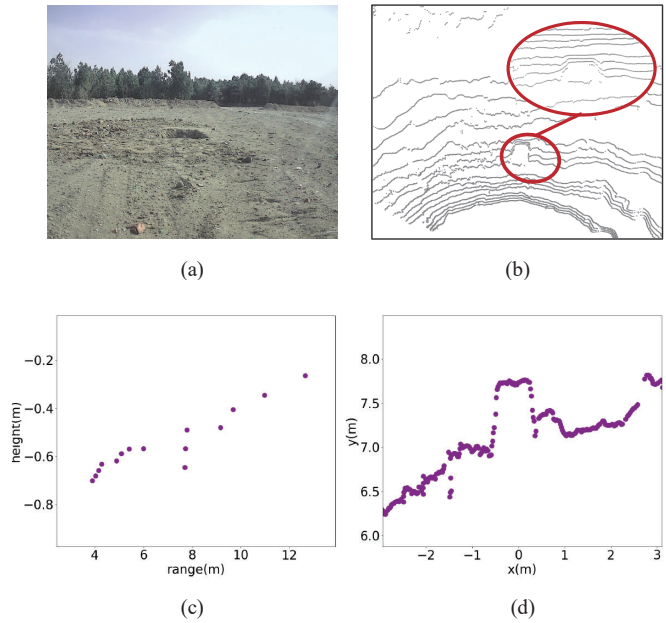

Fig 3. Schematic diagram of negative obstacle. (a) The negative obstacle in real scene. (b) The negative obstacle in the point cloud. (c) Point distribution in the radial direction when LiDAR scans the negative obstacle. (d) Point distribution of the laser beam as it scans a negative obstacle in one cycle.

After analyzing the LiDAR sensor sampling principle, we summarize the geometric features that can characterize negative obstacles in the 3D sparse point cloud as follows:

A. Distance distortion of adjacent scan points

Along the radial scanning direction of the LiDAR, when all the scanning points hit the ground, the radial distance of the scanning points increases with the inclination angle of the laser emitter. However, as shown in Fig 3 (c), when the scan point hits the trailing edge of the negative obstacle, the difference between the radial distance of the scan point and the previous scan point becomes large, and the difference between the radial distance of the scan point and the trailing edge of the scan point becomes small, causing a distortion in the original increase trend of radial distance.

B. Height sinking on trailing edge

Fig 3 (c) shows the result of one scan along the radial direction of the LiDAR. The scan points on the plane are distributed linearly, and when the scan points are located at the trailing edge of a negative obstacle, the height of these points is significantly less than the height of the surrounding points.

C. Local convexity of ring-shaped scanning

From the previous discussion, it is clear that the data obtained after each laser beam completes the scan cycle is distributed in a ring shape. Consider that when a scanning point hits the trailing edge of a negative obstacle, there is a significant increase in the measured distance of the point compared to the measured distance of the point hitting the ground. Therefore, when a particular laser beam scans over a negative obstacle, its ring-shaped data distribution produces a convexity of a certain width, as shown in Fig 3(d). This feature can effectively characterize the edges of negative obstacles in sparse point clouds.

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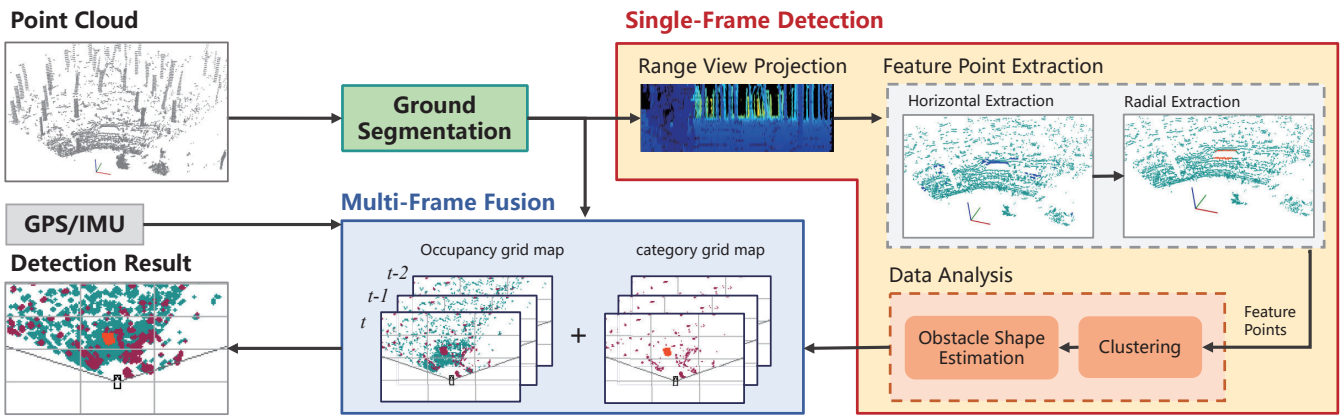

Fig 4. Overview of the proposed method. The output of the Negative Obstacle Detection module is a grid map where each cell contains positive or negative obstacle attributes. Then, the GPS/IMU information is utilized to fuse the historical grid map with the grid map at the current time t to output the final obstacle detection results.

IV. Method

A. The Overall Architecture

Fig 4. shows the overall architecture of our proposed method. The input sensor data consists of one LiDAR and GPS/IMU. The ground segmentation part is used to segment the point cloud into the ground and positive obstacles.

The single-frame detection part mainly consists of the following steps: 1) Projects the point cloud into a multi-channel range view image. 2) Combined with the segmentation results, row traversal and column traversal are performed on the image to calculate the negative obstacle geometric features between neighboring points, respectively, and the negative obstacle feature points are extracted after the comprehensive judgment of the features. 3) Clustering the feature points and statistical information about the features of each cluster. According to the statistical information, negative obstacle clusters are extracted, and shape estimation is performed for these clusters.

The multi-frame fusion part rasterizes the ground segmentation results and negative obstacle detection results into two grid layers, each containing different attributes of the detection results. Then, multi-frame fusion is performed on each grid layer separately to update the attribute information of each grid using Bayes' rule. Finally, the information of the two grid layers is combined to output continuous and stable detection results.

B. Ground Segmentation

Negative obstacles are usually located under the ground. Therefore, to reduce the negative obstacle detection complexity, the laser points that hitting on the positive obstacles should be filtered out. Fig 5. illustrates the ground segmentation method used in this work.

The point cloud is rasterized into multiple grid map layers. The variance layer G_{var} contains the z-axis (height direction) point coordinate variance for each cell. The min-z G_{minz} layer, max-z layer G_{maxz}, and point count layer G_{count} contain the minimum point's height value, maximum point's height value and point count for each cell, respectively. The variance of each cell is calculated using Welford's online algorithm [12]

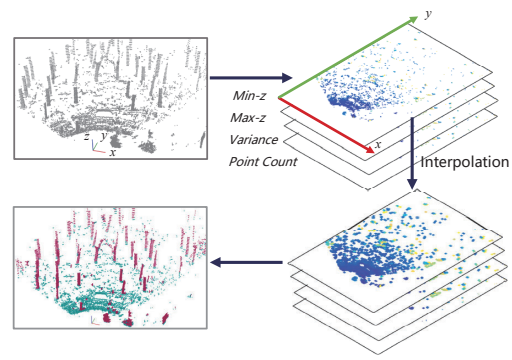

Fig 5. Flowchart of the ground segmentation method.

so that all grid map layers can be obtained by iterating through the point cloud once.

Next, we used a normalized convolution [13] to interpolate the min-z grid map to obtain the ground evaluation value for each cell.

Finally, we combine all the grid map layers to calculate the feature for each cell to segment the point cloud. If the feature in the cell meets the criteria defined as (1), we conclude all the points in this cell are positive obstacle points, and put these points into the set O_{positive}, where O_{positive} is the set of positive obstacle points.

$$\begin{cases} G_{\text{var}}(w,h) > T_{\text{var}} \\ (G_{\text{maxz}}(w,h) - G_{\text{elevation}}(i,h)) > T_{\text{height}} \\ G_{\text{count}}(w,h) > \dfrac{1}{r_{i,j} \cdot \Delta\delta \cdot \Delta\theta} \\ h \in [1,H], w \in [1,W] \end{cases} \tag{1}$$

Where w, h is the indices of cell in grid map, H is the height of the grid map, W is the width of the grid map, T_{var} is the threshold for the variance, T_{height} is the threshold for the height difference in each cell, $r_{w,h}$ is the distance between the cell (w, h) and the cell where the sensor is located, $\Delta\delta$ is the resolution of the grid map, and $\Delta\theta$ is the horizontal angular resolution of LiDAR.

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In addition, if the cell in the grid map does not meet the criteria by(1), we segment the point in this cell according to (2) .

$$\begin{cases} p_{i,j} \in O_{\text{positive}} & z_{i,j} - G_{\text{elevation}}(i,j) > T_{\text{ground}} \\ p_{i,j} \notin O_{\text{positive}} & z_{i,j} - G_{\text{elevation}}(i,j) \le T_{\text{ground}} \end{cases} \tag{2}$$

Where $p_{i,j}$ is the point that rasterized into grid cell (i, j), $z_{i,j}$ is the height value of $p_{i,j}$, T_{ground} is a threshold.

C. Single-frame Detection

1) Range View Projection:

To extract the negative obstacle geometry features, we project the point cloud into the range view for serialized representation. The LiDAR point (x, y, z) in the Cartesian coordinate system is converted to (r, θ, φ) in the Spherical coordinate system by (3), which is used to generate the range view image.

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \operatorname{atan2}(y, x) \\ \varphi = \operatorname{atan2}\left(z, \sqrt{x^2 + y^2}\right) \end{cases} \tag{3}$$

Where r is the range of the point, θ is the azimuth angle, φ is the inclination angle, as shown in Fig 6.

As shown in Fig 6, a range view image $I \in \mathbb{R}^{N \times M}$ represents the Spherical coordinate system in matrix form. The number of rows N of the image is equal to the number of LiDAR beams N_{beams} . The M column of the range view image covers the sensor's azimuth scanning angle range. Let (m, n) be the column index and row index of the image, respectively. Given a horizontal angular resolution $\Delta\theta$, the column index value m of point (r, θ, φ) in the image is $m = (\theta) / \Delta\theta$. The row index value n of point (r, θ, φ) is the index value of φ in the LiDAR inclination distribution.

Then, we map all LiDAR points to the image by computing the (m, n) of each point and assign four-dimensional data to each cell in the image as follows:

$$I_{m,n} = [p_{m,n}, r_{m,n}^{xy}] \; m \in [1, M], n \in [1, N] \tag{4}$$

Where $p_{m,n}(x_{m,n}, y_{m,n}, z_{m,n})$ is the LiDAR point in the Cartesian coordinate system that maps to the image cell (m, n) , and $r_{m,n}^{xy} = \sqrt{(x_{m,n})^2 + (y_{m,n})^2}$

Finally, the range view image constructed in this work is a matrix of size $N \times M \times 4$.

2) Candidate Point Extraction

Horizontal feature extraction: As discussed in Section II, we first Iterate through each row of the range view image to extract local convexity points.

In order to extract these feature points, we fit a quadratic polynomial to the radial distance $r_{m,n}^{xy}$ data contained in each

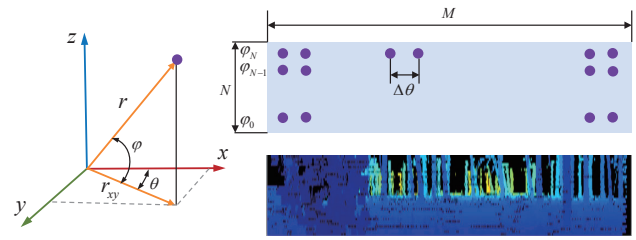

Fig 6. The range view image of the point cloud. (a) The relationship between the Cartesian coordinate system and the Spherical coordinate system. (b) The range view image mode. (c) An real example of the range view image (blue/yellow indicates the near/far radial distance of the point).

row of the image, and the quadratic polynomial function in row n is constructed as follows:

$$\begin{cases} f_n(m) = a_{n,0} + a_{n,1} \cdot \theta + a_{n,2} \cdot \theta^2 + a_{n,3} \cdot \theta^3 \\ \theta = m \cdot \Delta\theta + 30^\circ \end{cases} \tag{5}$$

Where $a_{n,0}, a_{n,1}, a_{n,2}, a_{n,3}$ are coefficients.

As shown in Fig 7, the difference between the fitting radial distance $f_n(m)$ and the real radial distance $r_{m,n}^{xy}$ determines whether the point $p_{m,n}$ is a local convexity point or not:

$$\begin{cases} p_{m,n} \in P_{\text{convex}} & \left(r_{m,n}^{xy} - f_n(m)\right) > T_{\text{convex}} \\ p_{m,n} \notin P_{\text{convex}} & \left(r_{m,n}^{xy} - f_n(m)\right) \le T_{\text{convex}} \end{cases} \tag{6}$$

Where P_{convex} is the set of local convexity points, and T_{convex} is the threshold we use.

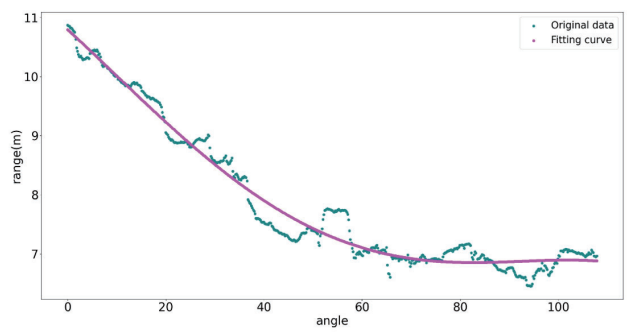

Fig 7. Fitting result of the data in one row of the range view image.

Notably, we only use points that are not part of the set of positive obstacle points O_{positive} obtained from the ground segmentation section.

Radial feature extraction: After obtaining the local convexity points, we traverse the range view image column-by-column to extract the final candidate negative obstacle points by computing the distance distortion and height sinking features of each point.

According to the analysis in Section II, if the point $p_{m,n}$ is consistent with (7), we can classify the point $p_{m,n}$ as the trailing edge point of the negative obstacle and put it into the

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set P_{trailing} ,where the P_{trailing} is the set of candidate trailing edge points, $P_{\text{trailing}} \subseteq P_{\text{convex}}$.

$$\begin{cases} 0 < r_{m,k}^{xy} - r_{m,n}^{xy} < T_{\text{front}} \\ T_{\text{nmin}} < r_{m,n}^{xy} - r_{m,v}^{xy} < T_{\text{nmax}} \\ 0 \le z_{m,k} - z_{m,n} \le T_{\text{edge}} \\ z_{m,v} - z_{m,n} \ge 0 \\ p_{m,n} \in P_{\text{convex}} \end{cases} \tag{7}$$

Where k is the indices of the nearest valid point to point $p_{m,n}$ forward from point $p_{m,n}$, v is the indices of the nearest valid point to point $p_{m,n}$ backward from point $p_{m,n}$, T_{front} , T_{nmax} , T_{nmin} and T_{edge} is the threshold.

If $p_{m,n}$ is classified as a trailing edge point, the adjacent point $p_{m,v}$ in front of the $p_{m,n}$ corresponds to the leading edge of the negative obstacle. Then, we extract all the $p_{m,v}$ and $p_{m,n}$ in the point cloud and form a set of negative feature point pairs:

$$P_{\text{neg_pair}} = \left\{ (p_{m,v}, p_{m,n}) \mid p_{m,n} \in P_{\text{trailing}} \right\} \tag{8}$$

3) Data Analysis

After obtaining the candidate's negative obstacle points, we analyze them to filter out the invalid points.

First, we cluster the points in P_{convex} into physically meaningful groups using the RBNN [14] cluster algorithm. The algorithm determines whether two points are the same object by calculating the Euclidean distance d between them. If d is less than a threshold T_d , the algorithm clusters the two points into the same object. We set T_d in this work as follows:

$$T_d = r_{m,n} \frac{\sin \Delta\theta}{\sin\left(\lambda - \Delta\theta\right)} + \sigma_d \tag{9}$$

Where λ is an angle threshold for determining the points to be of the same object, δ_d is a distance threshold for reducing the effect of measurement noise on clustering.

Then, the feature information in each cluster is extracted, which includes the number of trailing edge points N_{trailing} and the width of the clustering W_c . When the feature information of a cluster satisfies the following conditions, the cluster is determined to be a negative obstacle:

$$\begin{cases} N_{\text{trailing}} > T_{\text{ob_num}} \\ W_c > T_w \end{cases} \tag{10}$$

Where the $T_{\text{ob_num}}$ and T_{w} is the threshold.

Furthermore, we combine the points in the cluster that belong to $P_{\text{neg_pair}}$ and P_{convex} together and use these points to compute a convex hull polygon for each cluster.

D. Multi-frame fusion

Influenced by the sparsity of the point cloud, the single-frame detection approach may have missed detection. During the UGV 's movement, the large inclination angle of the point cloud's neighboring laser emitters results in a large scanning distance between the two laser beams, making some frames of the point unable to scan negative obstacles.

In order to improve the robustness of the negative detection, we apply the multi-frame fusion method to fuse the detection result from different frames.

1) Data rasterization

We rasterize all the ground points, positive points, and negative points into two grid map layers. The first grid map is filled with occupancy probability information for each cell and is used to maintain the obstacle detection results. The second grid map of each cell contains the obstacle category probability information for resolving the case of conflicting detection categories within the same cell. The size and the resolution of both grid maps are the same as the grid map that is mentioned in the ground segmentation section. For the probability occupancy grid map, there are three kinds of cell attributes $\{Occupancy, Free, Unknown\}$, and when the cell contains either positive or negative obstacles, the attribute of the cell is set to $Occupancy$. There are also three states for each cell in the probabilistic category grid map, which include $\{Positice, Negative, Null\}$.

2) Coordinate transformation

The GPS/IMU information is used to transform the previous frame grid map coordinate system F_{pre} to the current grid map coordinate system F_{cur} to update the attributes of each cell.

The coordinate transformation involves three coordinate systems, including the Universal Transverse Mercator Grid System (UTM), the vehicle coordinate system, and the grid map coordinate system, which is shown in Fig 8. The transformation between the vehicle coordinate system and the grid map coordinate system can be calculated by (11).

$$\begin{bmatrix} \dfrac{x_t^{map}}{\gamma} \\ \dfrac{y_t^{map}}{\gamma} \end{bmatrix} = \left(\begin{bmatrix} x_t \\ y_t \end{bmatrix} + \begin{bmatrix} x_v \\ -y_v \end{bmatrix} \right) \tag{11}$$

Where $\left[x_t^{map}, y_t^{map} \right]^T$ is the coordinate in the grid map, $\left[x_t, y_t \right]^T$ is the coordinate in the vehicle coordinate system, $\left[x_v, y_v \right]^T$ is the coordinate of the origin of the vehicle coordinate system in the grid map, γ is the resolution of the grid map.

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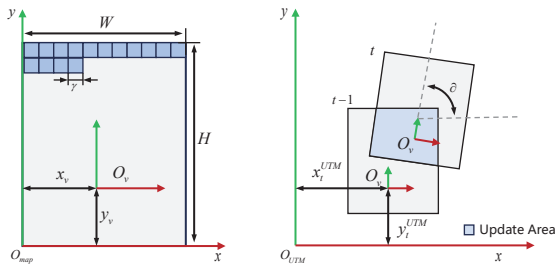


(a) (b)

Fig 8. Schematic diagram of the inter-transformation relationship between the three coordinate systems. O_v, O_{map} and O_{UTM} are the origin of the vehicle coordinate system, the grid map coordinate system and the UTM, respectively.

Based on the UTM coordinate system, the vehicle coordinate system transformation between time *t* and *t-1* can be calculated by (12).

$$\begin{cases} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = R_t \left(R_{t-1}^{-1} \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} x_{t-1}^{UTM} \\ y_{t-1}^{UTM} \end{bmatrix} - \begin{bmatrix} x_t^{UTM} \\ y_t^{UTM} \end{bmatrix} \right) \\ R_t = \begin{bmatrix} \sin(\partial) & \cos(\partial) \\ -\cos(\partial) & \sin(\partial) \end{bmatrix} \end{cases} \tag{12}$$

Where ∂ is the heading angle of the vehicle, ∂ is along the direction of x-axis in the UTM, rotate counterclockwise around z-axis, $\left[x_t^{UTM}, y_t^{UTM}\right]^T$ is the coordinate of the origin of the vehicle coordinate system in the UTM.

In this work, we first transform the data in the coordinate system of the grid map at time *t-1* to the vehicle coordinate system. The equations are then used to transform the data to the vehicle coordinate system at the time *t*. Finally, the inverse of (11) is utilized to transform the data in the vehicle coordinate system to the coordinate system of the grid map at time *t*.

3) Grid map update

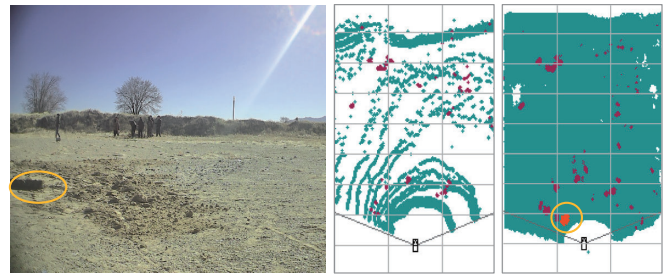
We use a binary Bayesian filter [15] to update the probability of each cell in the grid map.

Define the probability of cell $m(x_m, y_m)$ in the grid map at time *t as* $p(m_t \mid z_{1:t})$, and $z_{1:t}$ is the observation of continuous detection results from time *1* to time *t*. Then, the attribute probability in each cell can be updated by (13).

$$\begin{cases} S_t = \dfrac{p(m_{t-1} \mid z_{1:t-1}) \cdot p(m_{t-1} \mid z_t)}{\left(1 - p(m_{t-1} \mid z_{1:t-1})\right) \cdot \left(1 - p(m_{t-1} \mid z_t)\right)} \\ p(m_t \mid z_{1:t}) = \dfrac{S_t}{1 + S_t} \end{cases} \tag{13}$$

Where z_t is the observation at time *t*.

The probability of different observation types in each cell is defined as follows:



(a) (b) (c)

Fig 9. Comparison of the single-frame negative obstacle detection result and the multi-frame fusion negative obstacle detection result. (a) Real-world scene. (b) Negative obstacles cannot be detected by single-frame detection because there are no points on negative obstacles in the point cloud. (c) Multi-frame fusion detection result. The detection result is circled in yellow.

$$\begin{cases} p_{occ}\left(m_t \mid z_t^{occ} = Occupancy\right) = \varsigma,\ p_{occ}\left(m_t \mid z_t^{occ} = Free\right) = 1 - \varsigma \\ p_{occ}\left(m_t \mid z_t^{occ} = Unknown\right) = 0.5 \\ p_{attr}\left(m_t \mid z_t^{attr} = Positive\right) = \tau,\ p_{attr}\left(m_t \mid z_t^{attr} = Negative\right) = 1 - \tau \\ p_{attr}\left(m_t \mid z_t^{attr} = Null\right) = 0.5 \\ \tau > 0.5, \quad \varsigma > 0.5 \end{cases} \tag{14}$$

Where the $p_{occ}\left(m_t \mid z_t^{occ}\right)$ is the probability of the observation z_t^{occ} in the occupancy grid map at time *t*, the $p_{attr}\left(m_t \mid z_t^{attr}\right)$ is the probability of the observation z_t^{attr} in the category grid map at time *t*.

In the process of updating the grid map, for the category grid map, the observation z_t^{attr} in m_t would be set to *Null* if the observation z_t^{occ} in m_t is *Free*.

After the update method of the grid map and the probability of each observation type are defined. The final detection result can be obtained from the fusion grid map at time t by (16), which is shown in Fig 9.

$$\begin{cases} m_t^{fus} = Negative \quad p_{occ}\left(m_t \mid z_{1:t}^{occ}\right) < 0.5 \ \&\ p_{attr}\left(m_t \mid z_{1:t}^{attr}\right) > 0.5 \\ m_t^{fus} = Positive\ p_{occ}\left(m_t \mid z_{1:t}^{occ}\right) > 0.5 \ \&\ p_{attr}\left(m_t \mid z_{1:t}^{attr}\right) > 0.5 \end{cases} \tag{15}$$

Where the m_t^{fus} is the final detection result of the cell (x_m, y_m) in the grid map.

V. Experiments

A. Dataset

We carry out the experiments on the data we collected in real scenario tests. The dataset contains LiDAR data, RGB images and GPS/IMU data. The LiDAR data is collected by a C-Fans-32 which is horizontally mounted on top of the UGV. The dataset contains 3045 frames of data covering a variety of off-road scenarios such as dirt roads and woods, each containing typical negative obstacles, as shown in Fig 10. In addition, to evaluate the performance of the proposed method, the negative obstacles in each frame were hand-labeled as the ground truth.

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Fig 10.Typical negative obstacles in off-road environments.

B. Evaluation metrics

To quantify the results of the experiment, we projected the ground truth and the detection results into a grid map and calculated the area where the two intersect in the grid map. The size of the grid map is the same as the ground segmentation part in section Ⅲ. If the ratio of the intersected area to the true area is greater than 50%， the detection result would be positive. In this paper, precision and recall are used as indicators to judge the performance of negative obstacle detection. The precision and recall are calculated as follows:

$$Presicion = N_{TP} / \left(N_{TP} + N_{FP} \right) \tag{16}$$

$$Recall = N_{TP} / \left(N_{TP} + N_{FN} \right) \tag{17}$$

Where N_{TP} is the number of detection results of True Positive (TP), N_{FP} is the number of detection results of False Positive (FP), and N_{FN} is the number of detection results of False Negative (FN).

C. Parameter setting

For the ground segmentation section, the resolution of the grid map $\gamma = 0.2m$,height $H = 250$, width $W = 150$, and the vehicle position in the grid map $\left(x_v, y_v \right)$ is $(75, 25)$. For the other parameters in this section, we set $T_{\text{var}} = 1 \cdot e^{-5}$, $T_{\text{height}} = 0.2m$, $T_{\text{ground}} = 0.1m$. For the single-frame detection section, we set $T_{\text{convex}} = 0.3m$, $T_{\text{front}} = 0.1m$, $T_{\text{nmin}} = 0.5m$, $T_{\text{nmax}} = 5m$, $T_{\text{edge}} = 0.2m$, $T_{\text{ob_num}} = 10$ and $T_{\text{w}} = 0.5m$. For the multi-frame fusion section, we set $\varsigma = 0.72$, $\tau = 0.7$.

D. Results analysis

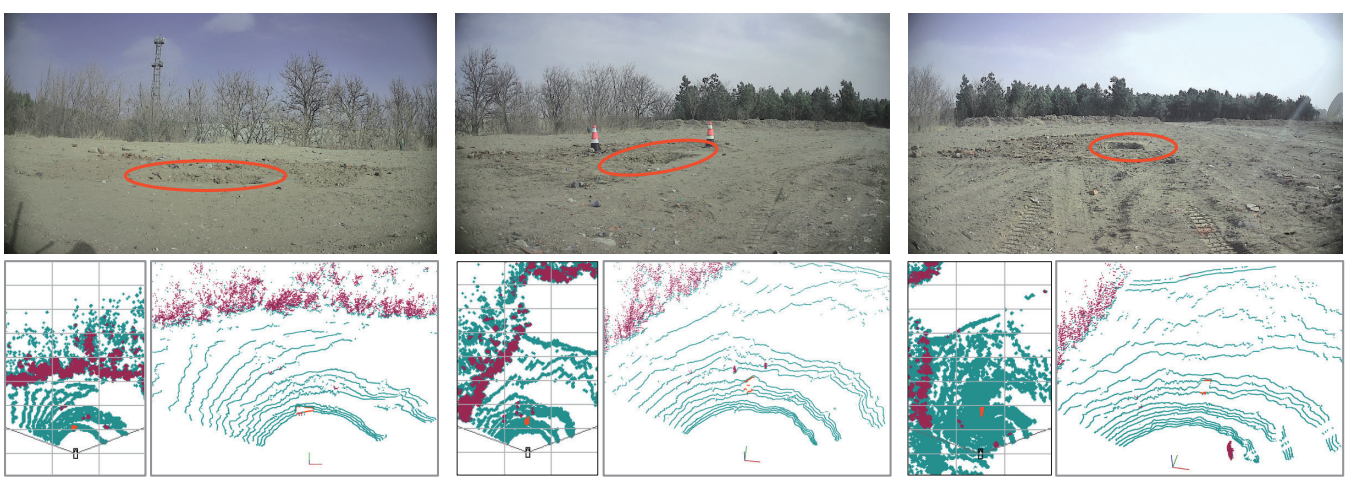
Fig 11. shows some representative negative detection results obtained from the proposed method. It can be seen that the method proposed in this paper is able to detect negative obstacles stably in off-road environments, even though the points hitting the negative obstacles are sparse.

Furthermore, the method proposed in this paper is quantitatively analyzed. To demonstrate the effectiveness of our method, we compare our method with the method proposed by [11]. As shown in TABLE I. , our proposed method has higher recall compared to the method [11], indicating a lower number of missed detections. This is because in some off-road scenarios with large terrain undulations, the UGV generates a large pitch angle when traveling, resulting in a large variation in the distance between adjacent scanning points of the LiDAR, and the features designed in method [11] do not describe this situation well, resulting in the inability to detect negative obstacles.

TABLE I. DETECTION RESULTS OF NEGATIVE OBSTACLES FOR DIFFERENT METHODS

Method	Precision	Recall
Proposed	98.1	**89.2**
[11]	**98.7**	69.5

In addition, we ran the proposed method in an environment with a 2.40GHz Intel Core i7-13700H CPU and 32GB of main memory, and it was able to achieve an average processing frame rate of over 142 Hz, providing a high level of real-time performance, which is shown in TABLE II. .



(a) (b) (c)

Fig 11.The negative detection results in off-road enviroments. (a) The detection result when the position of the negative obstacle is close to the sensor's detection blind zone and the leading edge features are not obvious. (b) The detection result when the edge of the negative obstacle is at an angle with the vehicle travelling direction. At this time, the points scanned at both the leading and trailing edges of the negative obstacle are sparse. (c) Negative obstacle detection result at a longer distance.

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TABLE II. COMPARISON OF THE RUNTIME

Method	Runtime(ms)
Proposed	**7.04**
[11]	15.30

VI. CONCLUSION

This study presents a negative obstacle detection method for sparse 3D LiDAR point clouds in off-road environments. This method uses three geometric features to describe negative obstacles by analyzing the horizontally mounted LiDAR scanning model. Then, negative obstacle points in a single-frame point cloud are extracted using a feature-based approach. In addition, a multi-frame fusion method based on a grid map is used to ensure the robustness of negative obstacle detection. Experiments were conducted in a variety of off-road environments containing typical negative obstacles. The experimental results verify that our work is able to detect negative obstacles in real time and stably.

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